The NoSAS score: a screening tool for sleep disordered breathing in the general population

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Research in context

Evidence before this study

Sleep disordered breathing (SDB) is a highly prevalent condition associated with neurocognitive deficits and increased cardiovascular risk. Diagnostic procedures such as respiratory polygraphy or polysomnography are time consuming and expensive, so they cannot be used as routine screening for SDB.

Added value of this study

We propose a new simple clinical score (NoSAS) to screen for SDB using clinical and polysomnographic data from a large population-based cohort. Using variables easily available in primary care practice, the NoSAS score allows to reliably rule out clinically significant SDB with a NPV of 90% and 98% in two ethnically different population-based cohorts.

Implications of all the available evidence

Compared to existing screening scores, NoSAS allows reducing the number of missed SDB diagnosis and the number of unnecessary nocturnal recordings.

ABSTRACT

Background The diagnosis of sleep-disordered-breathing requires overnight recordings such as polygraphy or polysomnography. Considering the costs and low availability of these procedures, it is recommended to pre-select patients at high risk. We aimed to develop a screening tool allowing the identification of individuals at-risk for sleep-disordered-breathing.

Methods 2121 subjects (48% men, 59±11 y.o.) from the population-based HypnoLaus cohort who underwent a clinical evaluation and a polysomnography at home were used to build a clinical score to identify subjects likely to have significant sleep-disordered-breathing. The score was externally validated in an independent sleep cohort (EPISONO, N=1042, 45% men, 42±14 y.o.). Its performance was compared to existing screening scores (STOP-Bang and Berlin).

Findings The NoSAS score allocates 4 points for Neck circumference >40 cm; 3 points for body-mass-index (BMI)≥25 and <30 kg/m^2; 5 points for BMI≥30 kg/m^2 (Obesity); 2 points for **S**noring; 4 points if Age>55 years and 2 points for men (Sex). Using a threshold of ≥8 points, NoSAS identified individuals at-risk for significant sleep-disordered-breathing with an area under the curve (AUC) of 0·74 (0·72-0·76). It showed an even higher performance in the EPISONO cohort: AUC=0·81 (0·77-0·85).The NoSAS score performed significantly better than STOP-Bang and the Berlin scores, which showed an AUC=0·67 (0·65-0·69) and 0·63 (0·61-0·66), respectively (both p-values <0·001).

Interpretation NoSAS is a simple, efficient and easy-to-implement score enabling the identification of individuals at-risk for sleep-disordered-breathing.

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Keywords: Nocturnal breathing disorder; sleep apnoea; apnoea-hypopnoea index; score; screening

INTRODUCTION

Sleep disordered breathing (SDB) is a highly prevalent disease characterized by repetitive collapse of the upper airway during sleep. Moderate-to-severe SDB has been shown to affect half of men and almost one quarter of women in the general middle to older age population.[1] SDB affects quality of life and different aspects of health domains,[2, 3] is associated with an increased incidence of cardiovascular events,[4, 5] hypertension,[6] diabetes,[7] metabolic syndrome,[8] and car accidents,[9] and represents an important public health issue. SDB diagnosis requires overnight recordings performed using portable limited channel recorders (respiratory polygraphy) or full polysomnography (PSG), either at home or in a sleep laboratory. These investigations are time consuming and expensive, and cannot be used as routine screening for SDB. Thus, different clinical scores such as the Berlin or the STOP-Bang have been previously proposed as screening tools. The Berlin questionnaire[10] includes information about snoring, daytime sleepiness or fatigue, obesity, and hypertension. It was created using a general clinical practice sample of 744 individuals of whom 100 underwent a polygraphic recording at home to confirm the diagnosis. The STOP-Bang score[11] combines information from a self-administered questionnaire about complaints of snoring, tiredness, observed apnoea, and high blood pressure (STOP) with clinical characteristics such as body mass index (BMI), age, neck circumference, and gender (Bang). It was created based on a large group of patients evaluated before a surgical procedure (n=2477) of whom 211 underwent a full PSG to confirm the diagnosis. These instruments are currently used in clinical practice, although they were developed in single specific clinical samples and validated using less sensitive (thermistor) recording technology (Berlin) or older respiratory events scoring criteria (STOP-Bang), which do not match current standards. Considering that these technical differences were shown to have a significant impact on SDB diagnostic and prevalence,[1] this study aimed to develop a new screening tool for SDB using current recording standards as reference and an external validation to ensure its reliability in different populations.

METHODS

Sample

The HypnoLaus Sleep Cohort study, described previously,[1] included a random subset of the population based CoLaus/PsyCoLaus cohort[12, 13] who had a full PSG at home between September 2009 and June 2013 in Lausanne, Switzerland. Sleep-related complaints and habits were investigated using the Pittsburgh Sleep Quality Index,[14] the Epworth Sleepiness Scale (ESS)[15] as well as the Berlin questionnaire[10] and the STOP-Bang score[11] for SDB. Clinical and demographic data were also collected (see supplementary material).

The ethics committee of the University of Lausanne approved the CoLaus/PsyCoLaus cohort study and the HypnoLaus Sleep Cohort study. We obtained written informed consent from all participants.

Polysomnography

Certified technicians equipped the subjects with a PSG recorder (Titanium, Embla®Flaga, Reykjavik, Iceland) between 5 and 8 pm at the Center for Investigation and Research in Sleep (CIRS, University Hospital of Lausanne, Switzerland). All sleep recordings took place in the patients' home environment and in accordance with the American Academy of Sleep Medicine (AASM) 2007 recommended setup specifications.[16] Breathing was recorded using nasal pressure sensors. Subjects currently under treatment for SDB (N = 38) were asked to discontinue their treatment one week prior to the sleep recording. Two trained sleep technicians blinded to the questionnaires' results scored manually the PSG recordings using Somnologica software (Version 5.1.1, Embla® Flaga, Reykjavik, Iceland). Each recording was reviewed by an expert sleep physician and a second sleep expert performed random quality checks. Quality control for concordance rate between the 2 PSG scorers was implemented periodically to ensure that both scorers achieved at least a 90% level of agreement for sleep stages and respiratory events and an 85% level of agreement for arousals.[17] Apnoea was defined as a drop of at least 90% of airflow from baseline, lasting 10 seconds or longer. Hypopnoeas were scored using the latest 2012 AASM criteria[18] (≥ 30% drop of airflow, lasting at least 10 seconds associated with either an arousal or a ≥ 3% O_2 saturation drop). The average

number of apnoeas and hypopnoeas per hour of sleep (apnoea-hypopnoea index [AHI]) was calculated. According to a recent paper by our group,[1] we defined significant SDB as an AHI > 20 events per hour.

Statistical analysis

Statistical analyses were conducted using R software (R Core Team, 2014).[19] To develop the screening score, based on demographic data and clinical characteristics, multiple factor analysis was used to visually explore the simultaneous relationships between continuous and categorical variables, and to identify SDB-related features. For each selected continuous variables, the threshold with the highest area under the curve were selected, using AHI > 20/AHI ≤ 20 as response variable and the continuous variable as predictor. The probability of significant SDB, AHI > 20 events per hour, was estimated using logistic regression for subjects with complete data and a score was created taking into account the contribution of each variable included in the logistic model: it was built as the sum of the assigned points, defined as the exponential of the estimated coefficient, rounded to the nearest integer. It was subsequently dichotomized into positive or negative. The performance of the score was then evaluated in terms of discrimination using the area under the curve (AUC) and the DeLong method[20] to calculate 95% confidence interval; positive predictive value (PPV), i.e., probability that subjects with a positive screening test truly have the disease; and negative predictive value (NPV), i.e., the probability that subjects with a negative screening test truly do not have the disease. In order to avoid over-fitting due to the use of the same data for estimation and evaluation, we used k-fold cross validation technique,[21] with k equal to ten. Briefly, this technique randomly splits the data into k equal size subsamples. It uses k-1 subsamples for the estimation and the remaining one for validation. The process is then repeated k times, with each of the k subsamples used once as the validation data. The k results are then combined to produce a single estimation. Finally, we calculated the percentage of “unnecessary” sleep recordings, defined as false positive rate, and the percentage of “missed” SDB diagnosis or false negative rate, if individuals with positive scores were referred for sleep recordings.

External validation

An independent external validation was conducted using data from the EPISONO population-based cohort,[22] carried out in São Paulo, Brazil, in 2007. The details of the study design and methods were previously published.[23] Briefly, EPISONO survey used a probabilistic three-stage cluster sample of São Paulo inhabitants to represent the population according to gender, age, and socio-economic status. Face-to-face interviews and in-lab full-night PSG using a nasal cannula and thermistor were performed. An initial sample size of 1101 subjects were included from the four homogenous socioeconomic macro-regions of São Paulo as classified by the National Institute of Demographics and Statistics (IBGE) census. Five percent refused to undergo the PSG exam, leading to a final sample size of 1042 participants. Apnoeas were defined as a drop of at least 90% of airflow from baseline, lasting 10 seconds or longer. Hypopnoeas were defined as ≥ 50% drop of airflow, lasting at least 10 seconds associated with either an arousal or a ≥ 3% O_2 saturation drop. The performance of the NoSAS score as well as the Berlin and STOP-Bang scores were evaluated and compared to one another in the EPISONO sample.

Role of the funding sources

The Faculty of Biology and Medicine of Lausanne, the Lausanne University Hospital (CHUV), Leenaards Foundation, and Ligue Pulmonaire Vaudoise funded the salary of the technicians who did the sleep recordings. The Swiss National Science Foundation funded the statisticians and supported the initial CoLaus cohort. GlaxoSmithKline supported the initial CoLaus cohort and funded the polysomnography recorders. The funders of the study had no role in study design, data collection, data analysis, data interpretation, or writing of the report. The corresponding author had full access to all data in the study and had final responsibility for the decision to submit for publication.

RESULTS

Sample characteristics

Of the 3051 consecutive contacted subjects from the CoLaus/PsyCoLaus cohort, 2168 (71·1%) agreed to have a PSG at home. Among the 2168 subjects who had a full PSG at home, 2121 were included and 47 were excluded (41 due to a total sleep time < 4h and six due to technical problems). Compared with the whole CoLaus/PsyCoLaus cohort, individuals who underwent PSG were similar in terms of age, sex, BMI, ethnic origin, and were considered as representative of Lausanne's general population.[12] From the 2121 HypnoLaus participants included in this analysis, the mean age was 59 ± 11 y.o., 48% were men and 57% were overweight. Overall, they were slightly overweight (mean BMI 25·6 ± 4·1 kg/m^2) with a mean neck circumference of 36·9 ± 3·9 cm and a mean waist-to-hip ratio of 0·92 ± 0·07. Approximately one-quarter reported alcohol consumption and one-fifth currently smoked. The prevalence of hypertension, diabetes and metabolic syndrome was 41%, 10% and 30%, respectively. Two-thirds of the participants reported snoring. The prevalence of significant SDB (AHI>20/h) was 26% (Table 1).

Clinical score

Initially, all available demographic and clinical variables classically related to SDB, such as age, gender, BMI, height, weight, neck circumference, waist-to-hip ratio, alcohol use, smoking, snoring, hypertension, diabetes, metabolic syndrome, and Epworth score were analysed by multiple factor analysis in order to identify blocks of variables. Using a graphical tool provided by this approach and logistic regression models, we identify the most relevant variables: Neck circumference, Obesity, Snoring, Age and Sex (NoSAS score). This score, which ranges from 0 to 17, allocates 4 points if neck circumference > 40 cm; 3 points if BMI ≥ 25 and <30 kg/m^2; 5 points if BMI ≥ 30 kg/m^2; 2 points if snoring is reported; 4 points if age > 55 years and 2 points for men (Table 2).

Based on its ability to discriminate subjects with a significant SDB (i.e. AHI > 20 events per h), PPV and NPV, we chose eight as a threshold, so the score was defined as positive if it was greater than or equal to eight and negative if it was less than eight points (Figure 1). The performance of the score was AUC: 0·74 (95% confidence interval 0·72 to 0·76); PPV: 0·47 (0·44

to 0·51); NPV: 0·90 (0·88 to 0·92) (Table 3). The performance of NoSAS was also compared to Stop-Bang and Berlin scores using cut-offs of 5, 10, 15, 20, 25 and 30/h (figure 2 and supplementary Tables S1 and S2).

Validation and comparison with other scores

In addition to the internal cross validation, the NoSAS score was also validated in the independent population-based EPISONO cohort (45% men, mean age 42 ± 14 years, 60% overweight), where the performance of NoSAS in terms of AUC, PPV and NPV was 0·81 (0·77 to 0·85), 0·33 (0·28 to 0·38) and 0·98 (0·96 to 0·99), respectively.

Comparing the classification capability of NoSAS, STOP-Bang and Berlin scores in HypnoLaus subjects, NoSAS performed significantly better than the other scores in both cohorts: in HypnoLaus the AUC was 0·74 (0·72 to 0·76), 0·67 (0·65 to 0·69) and 0·63 (0·61 to 0·66) for NoSAS, STOP-Bang and Berlin score respectively, both p-values < 0·001. In EPISONO, the AUC were 0·81 (0·77 to 0·85), 0·68 (0·63 to 0·73) and 0·65 (0·59 to 0·70) for NoSAS, STOP-Bang and Berlin score respectively, both p-values < 0·001 (Table 3).

The percentage of correct SDB classification, i.e. positive NoSAS when AHI >20/h and negative NoSAS when AHI ≤ 20, in HypnoLaus and EPISONO were 72% and 78% for NoSAS, 54% and 65% for STOP-Bang, and 70% and 68% for Berlin (Table 4).

DISCUSSION

Using a large population-based cohort, we propose a new simple clinical score to screen for significant SDB. Our results show that, using variables easily available in primary care practice, the NoSAS score allows to reliably rule out clinically significant SDB with a NPV of 90 and 98% in two ethnically different population-based cohorts. Compared to existing screening scores, NoSAS allows reducing the number of unnecessary nocturnal recordings as well as the number of missed SDB diagnosis.

An ideal SDB screening score should have a high sensitivity to avoid false negative results but should also be specific enough to avoid referring low risk patients for costly and time consuming sleep recordings. The STOP-Bang score showed quite a poor discrimination

performance, with about two third of the population with a positive result (which yielded 45% of unnecessary sleep recordings) even though it had a low rate of missed diagnosis. The Berlin score also showed a poor discrimination in both cohorts and its sensitivity was quite low in our sample yielding the highest rate of missed diagnosis of the three scores, which is unfortunate for a screening tool. As suggested by the higher AUC and correct classification rate, the NoSAS score seems to represent the best sensitivity-specificity compromise allowing to reliably rule out significant SDB without yielding too many unnecessary sleep investigations.

Overall, the NoSAS score was able to detect 97·8% of subjects with severe SDB which gives a false negative rate 2·2%. Although the proportion of missed SDB with NoSAS using different thresholds is much lower than with Berlin questionnaire, it is higher than with STOP-Bang. This very low false negative rate of STOP-Bang score is however obtained at the cost of a very high false positive rate (55·6%). Moreover, the ability of STOP-Bang to correctly classify subjects for a cut-off of 20/h is close to a random classification (53·8%) and worse than random for severe SDB (44%), which questions its utility in these SDB categories.

Interestingly, almost all false negative subjects with the NoSAS score had a neck circumference ≤ 40 cm, 35% had a BMI between 25 and 30 kg/m^2, and only 4% had a BMI ≥ 30 kg/m^2. We can thus speculate that the cause of their sleep breathing disorder is more likely related to maxillofacial deformities, to a higher loop gain or to upper airway muscle control dysfunction than to obesity.

The fact that the NoSAS performed consistently well in two unrelated samples with a different age range, different ethnic background and habits is an asset and strongly emphasizes its consistency and generalizability to different populations. Conversely, the STOP-Bang seems to be less consistent, as reported in a recent review,[24] since it showed a very low specificity to predict moderate-to-severe and severe SDB in different population samples.[25, 26] We also observed variations in the STOP-Bang and Berlin performances between the HypnoLaus and the EPISONO cohorts whereas the performance of NoSAS appeared to be more stable. This higher stability may be due to the lower number of subjective variables included in NoSAS.

In order to be time efficient, a screening score should include a limited number of items related to easily available and objective parameters. The NoSAS score uses biometric items such as age,

gender and BMI, which are part of any standard clinical assessment, and neck circumference which can be easily measured. As the aim of this analysis was to develop a reliable and easily applicable score, we tried to limit the number of subjective variables, such as witnessed sleep apnoeas which require the observation of a bed partner and subjective sleepiness which was not associated with SDB severity in our cohort.[1] This lack of association may be due to the multiple factors that influence reported sleepiness such as sleep duration, medications, mood disorders, or other sleep disorders. Also we did not include hypertension as it did not appear as a major independent predictor of SDB in our cluster analysis. This may also be due to the multiple factors that can influence blood pressure. The presence of snoring is the only subjective item we included in the NoSAS score because of its strong statistical association with clinically significant SDB. However, we did not include snoring severity and frequency since these parameters are more subjective and difficult to quantify.

Compared to the eight items of the STOP-Bang and the nine to eleven questions (depending on the "snore question") and the three separate calculations required for the Berlin questionnaire, we believe that the NoSAS scores, which includes only five items, is easier to use in clinical practice. In addition a free smartphone application will be available as soon as the paper is accepted for publication. We can thus hope that this easy screening score could stimulate general practitioners to screen for SDB and decrease the proportion of undiagnosed and untreated SDB patients.

One of the strength of our analysis is that it relies on a large population-based sample with more than 2000 polysomnographic recordings compared to 100 and 211 recordings used to validate the Berlin and STOP-Bang scores. Moreover, we used up-to-date recording techniques and scoring criteria currently used in sleep laboratories, which probably increases NoSAS relevance for current clinical practice. We also believe that the external validation in an unrelated cohort from another continent showing a high reproducibility of the NoSAS score is an asset.

There are also limitations to our analysis that need to be taken into account. First, we chose to an AHI >20/h as cut-off to build the score based on the results of the first analysis of HypnoLaus cohort showing significant associations between SDB and hypertension, diabetes

metabolic syndrome and depression almost exclusively in the upper quartile (AHI>20·6/h).[1] This decision was obviously arbitrary but the NoSAS score seems perform well also with other AHI cut-offs. When using thresholds of 5/h, 10/h, 15/h, 20/h and 30/h we saw that the performance of the NoSAS score remains higher than with the other scores for AHI thresholds ≥ 10/h. Since the currently recommended recording techniques and scoring criteria for respiratory events are highly sensitive yielding high SDB prevalence figures,[27] we believe that using an AHI threshold <10/h would not be clinically relevant. Second, there was some missing data for the calculation of each of the scores due to participants who did not fill the questionnaires properly. This proportion was, however, rather low in general and we do not believe that it had a significant impact on the results because we are not in the framework of a point estimate like a treatment or an exposure effect, for instance. Finally, the NoSAS score was built and validated in population-based cohorts which may differ from clinical populations. This score will thus have to be further validated in specific clinical samples such as pre-surgical patients for example.
In conclusion, NoSAS is a simple and easy to interpret SDB screening score, which could help clinicians to better identify individuals at risk for significant SDB and decrease the rate of missed diagnosis.

Contributors

HMS, JHR, RH wrote the manuscript. HMS, CH, SBT were responsible for analysis. HMS, JHR and RH were responsible for the interpretation of results. HMS, CH, PMV, PV, GW, MP, MT, SBT, LB, ST, JHR and RH were involved in acquisition of data. All authors critically revised the manuscript for important intellectual content. JHR and RH supervised the study and are the guarantors.

Declaration of interests

The authors declared no conflict of interest.

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REFERENCES

1. Heinzer R, Vat S, Marques-Vidal P, et al. Prevalence of sleep-disordered breathing in the general population: the HypnoLaus study. *The lancet Respiratory medicine*. 2015; 3: 310-8.
2. Jordan AS, McSharry DG and Malhotra A. Adult obstructive sleep apnoea. *Lancet*. 2014; 383: 736-47.
3. Malhotra A and White DP. Obstructive sleep apnoea. *Lancet*. 2002; 360: 237-45.
4. Marin JM, Carrizo SJ, Vicente E and Agusti AG. Long-term cardiovascular outcomes in men with obstructive sleep apnoea-hypopnoea with or without treatment with continuous positive airway pressure: an observational study. *Lancet*. 2005; 365: 1046-53.
5. Yaggi HK, Concato J, Kernan WN, Lichtman JH, Brass LM and Mohsenin V. Obstructive sleep apnea as a risk factor for stroke and death. *The New England journal of medicine*. 2005; 353: 2034-41.
6. Peppard PE, Young T, Palta M and Skatrud J. Prospective study of the association between sleep-disordered breathing and hypertension. *The New England journal of medicine*. 2000; 342: 1378-84.
7. Kendzerska T, Gershon AS, Hawker G, Tomlinson G and Leung RS. Obstructive sleep apnea and incident diabetes. A historical cohort study. *American journal of respiratory and critical care medicine*. 2014; 190: 218-25.
8. Coughlin SR, Mawdsley L, Mugarza JA, Calverley PM and Wilding JP. Obstructive sleep apnoea is independently associated with an increased prevalence of metabolic syndrome. *European heart journal*. 2004; 25: 735-41.
9. Teran-Santos J, Jimenez-Gomez A and Cordero-Guevara J. The association between sleep apnea and the risk of traffic accidents. Cooperative Group Burgos-Santander. *The New England journal of medicine*. 1999; 340: 847-51.
10. Netzer NC, Stoohs RA, Netzer CM, Clark K and Strohl KP. Using the Berlin Questionnaire to identify patients at risk for the sleep apnea syndrome. *Annals of internal medicine*. 1999; 131: 485-91.
11. Chung F, Yegneswaran B, Liao P, et al. STOP questionnaire: a tool to screen patients for obstructive sleep apnea. *Anesthesiology*. 2008; 108: 812-21.
12. Firmann M, Mayor V, Vidal PM, et al. The CoLaus study: a population-based study to investigate the epidemiology and genetic determinants of cardiovascular risk factors and metabolic syndrome. *BMC cardiovascular disorders*. 2008; 8: 6.
13. Preisig M, Waeber G, Vollenweider P, et al. The PsyCoLaus study: methodology and characteristics of the sample of a population-based survey on psychiatric disorders and their association with genetic and cardiovascular risk factors. *BMC Psychiatry*. 2009; 9: 9.
14. Buysse DJ, Reynolds CF, 3rd, Monk TH, Berman SR and Kupfer DJ. The Pittsburgh Sleep Quality Index: a new instrument for psychiatric practice and research. *Psychiatry research*. 1989; 28: 193-213.
15. Johns MW. A new method for measuring daytime sleepiness: the Epworth sleepiness scale. *Sleep*. 1991; 14: 540-5.
16. Iber C A-IS, Chesson A, and Qaun SF. The AASM Manual for the scoring of sleep and associated events : Rules, terminology and technical specifications. *1st ed: Westchester, Illinois: American Academy of Sleep Medicine*. 2007.
17. Redline S, Sanders MH, Lind BK, et al. Methods for obtaining and analyzing unattended polysomnography data for a multicenter study. Sleep Heart Health Research Group. *Sleep*. 1998; 21: 759-67.
18. Berry RB, Budhiraja R, Gottlieb DJ, et al. Rules for scoring respiratory events in sleep: update of the 2007 AASM Manual for the Scoring of Sleep and Associated Events. Deliberations of the Sleep Apnea

Definitions Task Force of the American Academy of Sleep Medicine. *Journal of clinical sleep medicine : JCSM : official publication of the American Academy of Sleep Medicine*. 2012; 8: 597-619.
19. R Core Team. R: A language and environment for statistical computing. Vienna, Austria: R Foundation for Statistical Computing,, 2014.
20. DeLong ER, DeLong DM and Clarke-Pearson DL. Comparing the areas under two or more correlated receiver operating characteristic curves: a nonparametric approach. *Biometrics*. 1988; 44: 837-45.
21. Stone M. Cross-Validatory choice and assessment of statistical predictions. *Journal of the Royal Statistical Society Series B* 1974; 36: 111-47.
22. Tufik S, Santos-Silva R, Taddei JA and Bittencourt LR. Obstructive sleep apnea syndrome in the Sao Paulo Epidemiologic Sleep Study. *Sleep medicine*. 2010; 11: 441-6.
23. Santos-Silva R, Tufik S, Conway SG, Taddei JA and Bittencourt LR. Sao Paulo Epidemiologic Sleep Study: rationale, design, sampling, and procedures. *Sleep medicine*. 2009; 10: 679-85.
24. Miller JN and Berger AM. Screening and assessment for obstructive sleep apnea in primary care. *Sleep medicine reviews*. 2015; 29: 41-51.
25. Silva GE, Vana KD, Goodwin JL, Sherrill DL and Quan SF. Identification of patients with sleep disordered breathing: comparing the four-variable screening tool, STOP, STOP-Bang, and Epworth Sleepiness Scales. *Journal of clinical sleep medicine : JCSM : official publication of the American Academy of Sleep Medicine*. 2011; 7: 467-72.
26. El-Sayed I. Comparison of four sleep questionnaires for screening obstructive sleep apnoea. *Egyptian Journal of Chest Diseases and Tuberculosis* 2012; 61: 433-41.
27. Heinzer R, Marti-Soler H and Haba-Rubio J. Prevalence of sleep apnoea syndrome in the middle to old age general population. *The Lancet Respiratory Medicine*. 2016; 4: e5-e6.

FIGURE LEGEND

Figure 1 Evaluation of the performance of the NoSAS score for each possible threshold. Results from the 10-fold cross validation method. AUC, area under the curve; PPV, positive predictive value; NPV, negative predictive value. The selected threshold was 8, leading to an AUC=0·74, a VPP=0·47 and a NPV=0·90.

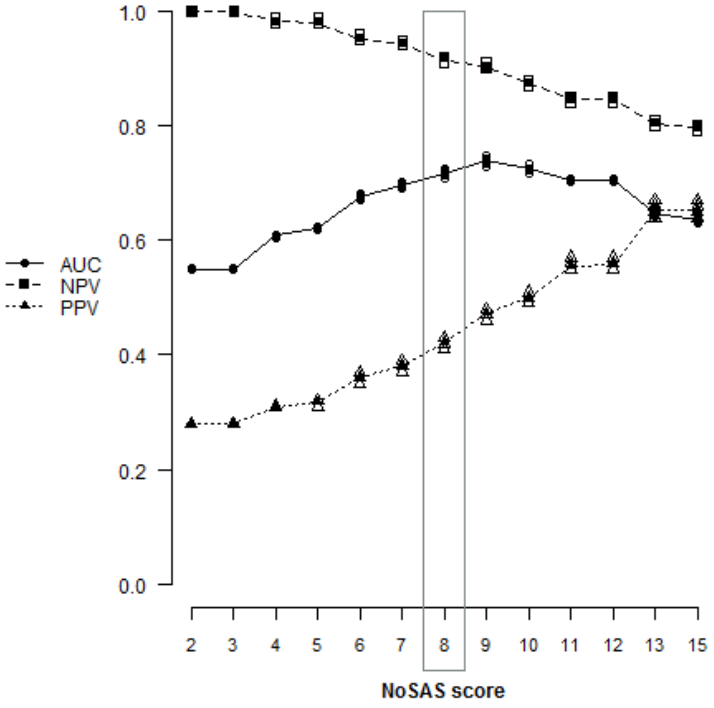
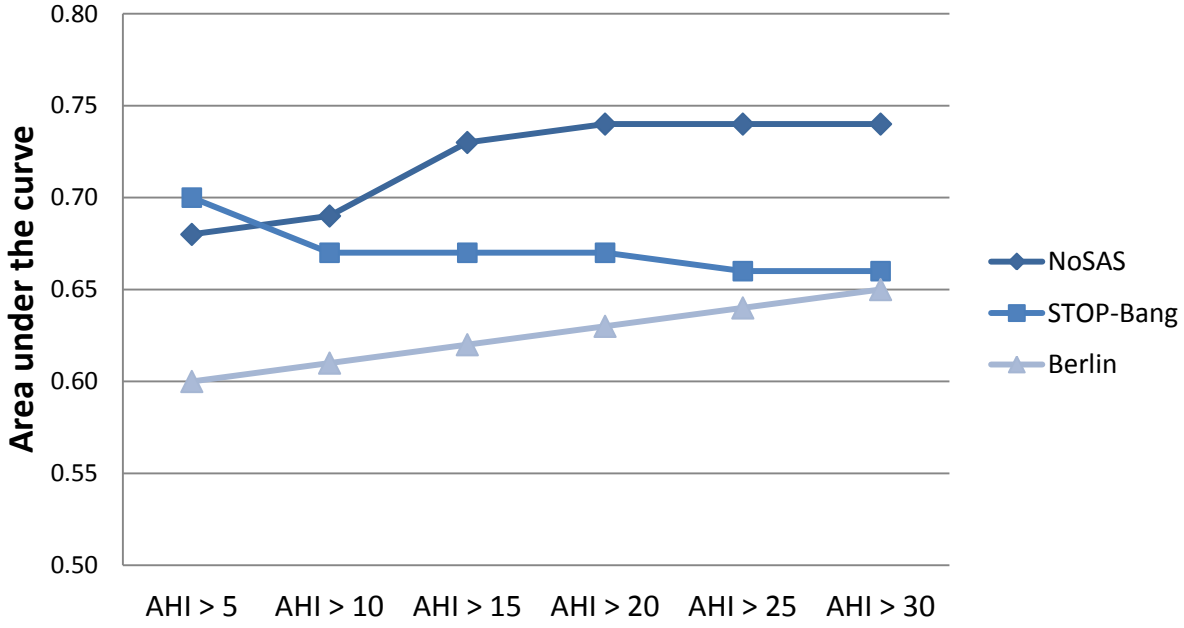


Figure 2 Performance of NoSAS compared to STOP-Bang and Berlin scores using cut-offs of 5, 10, 15, 20, 25 and 30/h (see also Supplementary tables S1 and S2).



TABLES

Table 1 Demographic data and characteristics of the global HypnoLaus cohort. Values represent raw count (%) unless otherwise specified.

	HypnoLaus cohort (N= 2121)
Age, years*	59 ± 11
Gender (men %)	795 (47·1)
Body mass index, kg/m^2*	25·6 ± 4·1
Neck circumference, cm*	36·9 ± 3·9
Waist-to-hip ratio*	0·92 ± 0·07
Alcohol use (%)	560 (26·0)
Current smokers (%)	395 (18·6)
Snoring (%)	1164 (54.9)
Hypertension (%)	877 (41·3)
Diabetes (%)	212 (10·0)
Metabolic syndrome (%)	641 (30·2)
Apnoea-hypopnoea index *>20/h*	551 (26·0)
Epworth score†	6 [3-9]
> 10	258 (12·2)
PSQI score†	4 [3-7]
Berlin score ≥ 2 (%)	525 (24·8)
STOP-Bang ≥ 3 (%)	1076 (69·0)

* mean ± SD; † Median [25th-75th percentiles]; PSQI : *Pittsburgh Sleep Quality Index. Arterial hypertension was defined as a systolic blood pressure (SBP) ≥140 mmHg and/or a diastolic blood pressure (DBP) ≥90 mmHg or current use of antihypertensive medication. Diabetes was defined as a fasting blood glucose level ≥ 7 mmol/L (126 mg/dL) or current use of antidiabetic medication. The metabolic syndrome (MS) was defined according to the Adult Treatment Panel III report (ATP-III). Drinking habits and the number of drinks taken during the evening preceding the polysomnography were self-reported. Waist-Hip Ratio was calculated as the ratio of the circumference of the waist to that of the hip as recommended by the World Health Organization (WHO). Neck circumference was measured between mid-cervical spine and mid-anterior neck*

just below the laryngeal prominence.

Table 2 Clinical score computation and decision rule for estimating moderate sleep disorder breathing in the general population: The NoSAS score.

Clinical features	NoSAS score
Neck circumference > 40 cm	4 points
Obesity	
BMI ≥ 25 kg/m^2 and < 30 kg/m^2	3 points
BMI ≥ 30 kg/m^2	5 points
Snoring	2 points
Age > 55 years	4 points
Sex: Male	2 points
High probability of significant sleep disordered breathing if NoSAS score ≥ 8	

BMI, body mass index

Table 3 Estimated performance (95% confidence interval) of different scores in the HypnoLaus cohort and external validation in the EPISONO cohort for identification of subjects with significant SDB (apnoea-hypopnoea index >20/h).

	Estimated prevalence	AUC	Sensitivity	Specificity	PPV	NPV
HypnoLaus						
NoSAS	43·3	0·74 (0·72-0·76)	0·79 (0·75-0·82)	0·69 (0·67-0·72)	0·47 (0·44-0·51)	0·90 (0·88-0·92)
STOP-Bang	69·0	0·67 (0·65-0·69)	0·94 (0·91-0·96)	0·40 (0·37-0·43)	0·35 (0·33-0·39)	0·95 (0·93-0·97)
Berlin	24·8	0·63 (0·61-0·66)	0·41 (0·37-0·45)	0·81 (0·79-0·83)	0·43 (0·39-0·47)	0·79 (0·77-0·81)
EPISONO						
NoSAS	11·4	0·81 (0·77-0·85)	0·85 (0·77-0·91)	0·77 (0·75-0·80)	0·33 (0·28-0·38)	0·98 (0·96-0·98)
STOP-Bang	11·6	0·68 (0·63-0·73)	0·72 (0·63-0·80)	0.64 (0·61-0·67)	0·21 (0·17-0·25)	0·95 (0·92-0·96)
Berlin	11·5	0·65 (0·59-0·70)	0·61 (0·52-0·70)	0·68 (0·65-0·71)	0·20 (0·16-0·24)	0·93 (0·91-0·95)

AUC: Area under de curve; PPV: positive predictive value; NPV: negative predictive value

Table 4 Percentage of correct classification, "unnecessary" sleep recordings and "missed" SDB diagnosis according to different scores in the HypnoLaus cohort and in the EPISONO cohort (external validation cohort)

HypnoLaus: Prevalence of SDB (AHI>20/h) by polysomnography, 26·1%						
	NoSAS (n=1703)		**STOP-Bang (n=1559)**		**Berlin (n=2115)**	
	-	**+**	**-**	**+**	**-**	**+**
AHI ≤ 20	871	388	460	697	1262	300
AHI > 20	95	349	23	379	328	225
% Correct classification	71·6%		53·8%		70·3%	
% Unnecessary sleep recordings	22·8%		44·7%		14·2%	
% Missed SDB diagnosis	5·6%		1·5%		15·5%	

EPISONO: Prevalence of SDB (AHI>20/h) by polysomnography, 11·5%						
	NoSAS (n=1042)		**STOP-Bang (n=1016)**		**Berlin (n=1036)**	
	-	**+**	**-**	**+**	**-**	**+**
AHI ≤ 20	714	208	572	326	628	290
AHI > 20	18	102	33	85	46	72
% Correct classification	78·3%		64·7%		67·6%	
% Unnecessary sleep recordings	20·0%		32·1%		28·0%	
% Missed SDB diagnosis	1·7%		3·2%		4·4%	

AHI: apnoea-hypopnoea index